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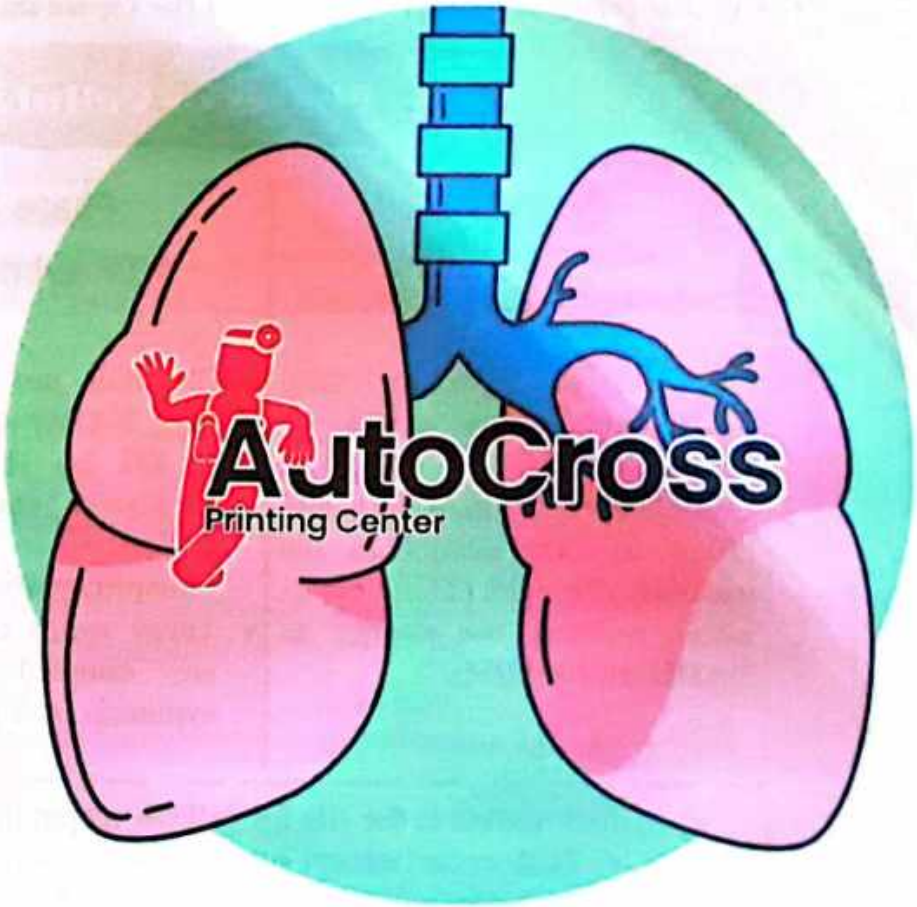
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BIOCHEMISTRY ACADEMY

Chapter

Glucose oxidation & Shuttles

Respiratory Module



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Cellular Respiration

- **Using O_2 and energy-derived from oxidizing fuel** (CHO, lipids & protein) to generate **ATP**.
- Cells require O_2 to generate ATP from the oxidation of fuels to CO_2 and water.
- Cellular respiration uses **over 90% of the O_2 we inhale.**

Phases of Cellular Respiration

Phase 1
Oxidation of fuels (carbohydrates, lipid & protein)

- Oxidation of (glucose, fatty acids, and amino acids) → **acetyl CoA.**
- Complete oxidation of the acetyl group to CO_2 occurs in the tricarboxylic acid (TCA) cycle, which **collects the energy as NADH and $FADH_2$.**

Phase 2
ATP synthesis

- Electrons are transferred from **NADH and $FADH_2$ to O2 by the electron transport chain (ETC)** Called **oxidative phosphorylation.**
- These series of reactions are coupled with the synthesis of ATP.

Phase 3
Consumption of ATP

ATP are used for processes such as:
- ✓ muscle contraction (**mechanical**)
- ✓ Na^+-K^+ pump (**transport**),
- ✓ Synthesis of larger molecules such as DNA in anabolic pathways (**biosynthetic**).
- ✓ Or detoxification (**biochemical**).

ATP is hydrolyzed to ADP + (Pi), or AMP + pyrophosphate (PPi).

- **Mitochondria** is the site for cellular respiration:-
  - ✓ TCA cycle "except succinate dehydrogenase" and oxidation of fatty acids located in the **mitochondrial matrix (phase 1).**
  - ✓ ETC are in the **Inner mitochondrial membrane (phase 2).**

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Examples of cellular respiration

- **Glucose** is a **universal fuel** used to generate ATP in every cell type in the body.
- **In glycolysis,** 1 glucose is oxidized to 2 pyruvate + 2 NADH by cytosolic enzymes + 2 ATP by substrate-level phosphorylation.

1. In anaerobic glycolysis

- NADH is reoxidized by conversion of pyruvate to **lactate**, enters the blood.
- Produce only **2 ATP.**
- Important for cells **lack mitochondria** as 'RBCS & kidney medulla' or tissues with **diminished blood flow** (exercising muscle).

2. In aerobic glycolysis

- NADH produced from glycolysis is reoxidized via the **electron transport chain, and the pyruvate is transported into the mitochondria** where it is oxidized to acetyl CoA which is the major fuel of the TCA cycle.

Fate of the end products of aerobic glycolysis (pyruvate and NADH)

A. Cytosolic NADH:

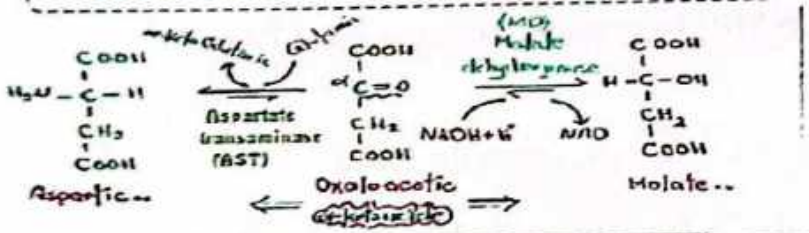
- Cytosolic NADH **cannot directly pass through inner mitochondrial membrane** to reach the electron transport chain.
- NADH is indirectly transferred to the inside of the mitochondria by what is called the shuttle mechanisms as **malate aspartate shuttle or Glycerol phosphate shuttle.**

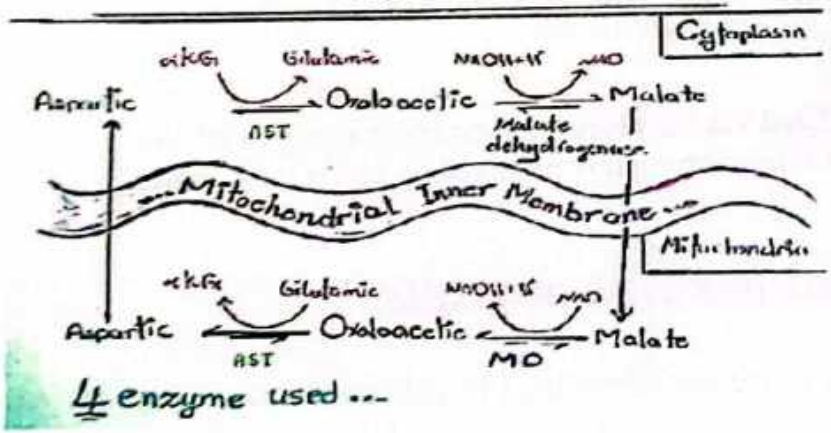
B. Pyruvate:

- In presence of O_2 and mitochondria, pyruvate is transported to mitochondria and undergoes **oxidative decarboxylation to acetyl CoA**

Shuttles of transfer NADH to Mitochondria

1. Malate shuttle (Malate-Aspartate)

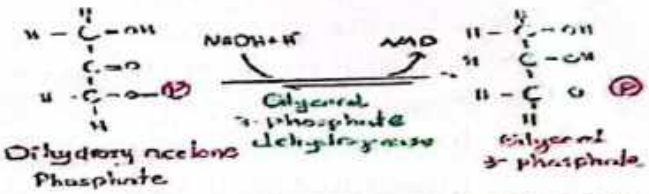


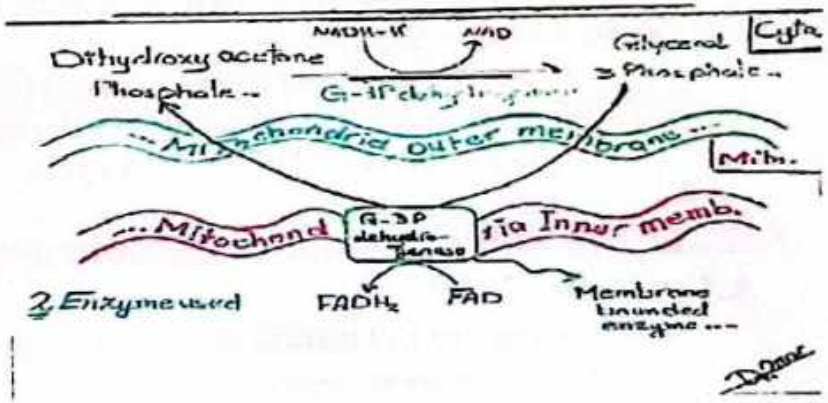


- Malate is transported to the mitochondria using **malate translocase**
- Oxaloacetate **can't** cross inner mitochondrial membrane.
- Aspartate is transported out to the cytosol using an **aspartate/glutamate exchange translocase.**
- NADH produce **3 ATP.**

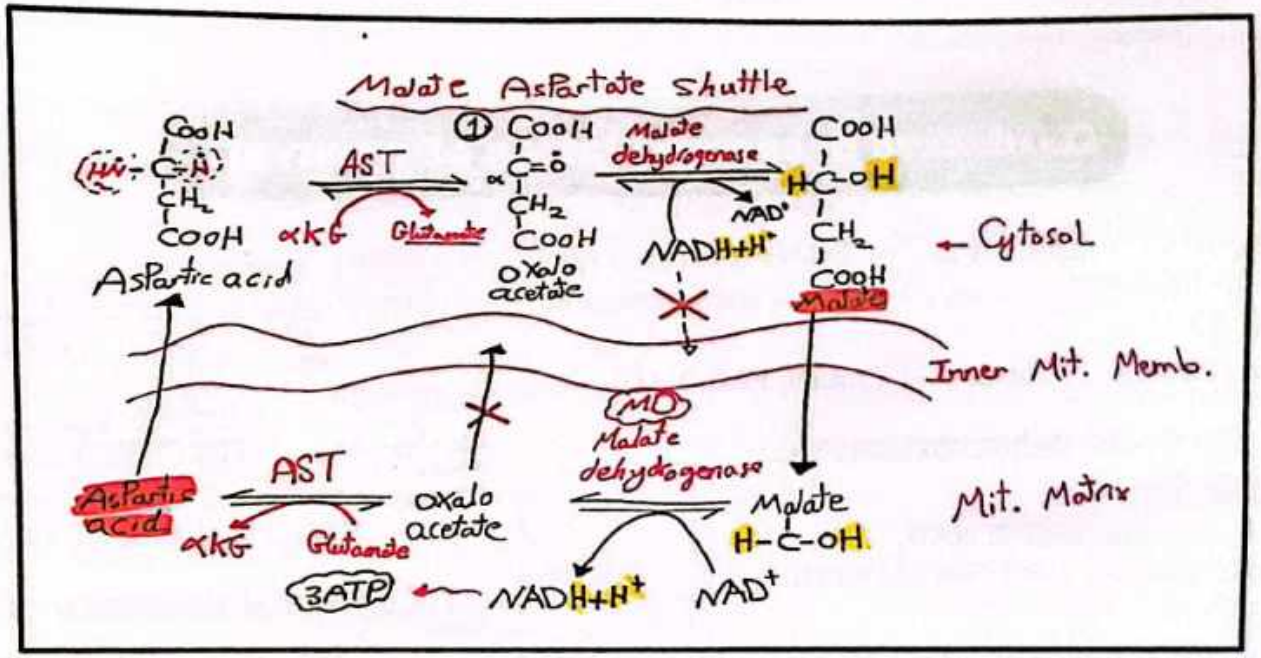
2. Glycerol phosphate shuttle

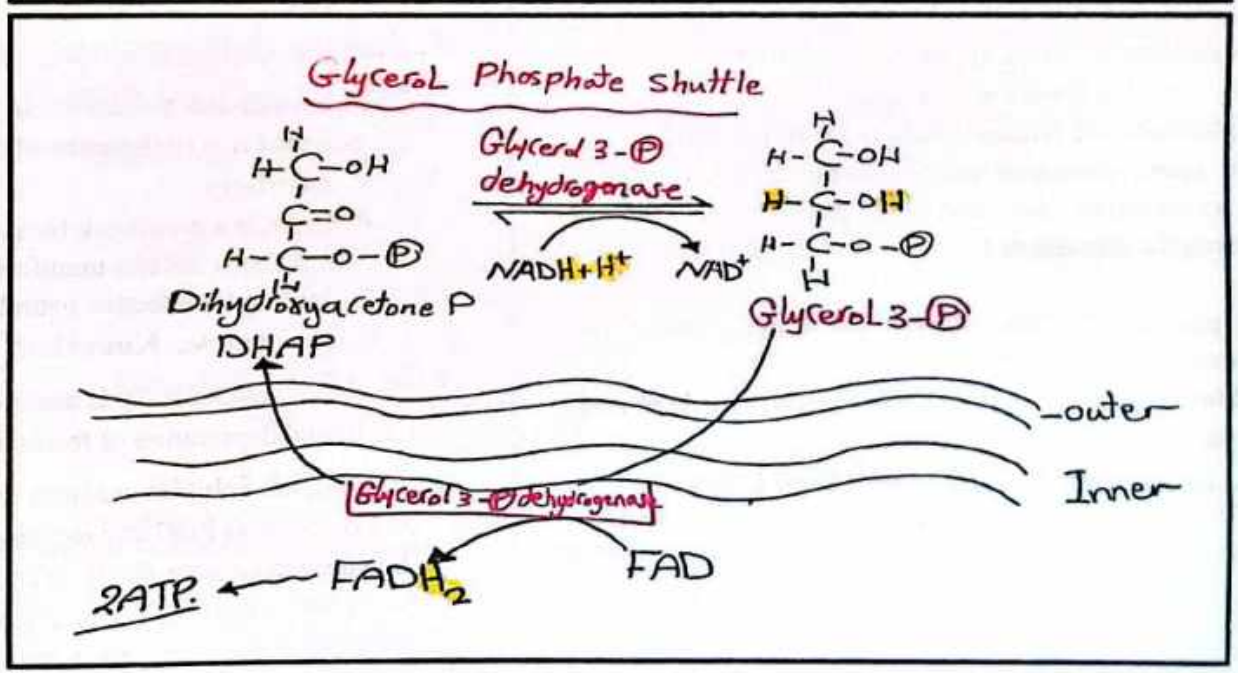
The major shuttle in most tissues





- Glycerol-3P **diffuse** to the inner mitochondrial membrane.
- Electrons are donated to FAD forming **$FADH_2$** → produce **2 ATP.**
- **DHAP return to cytosol.**

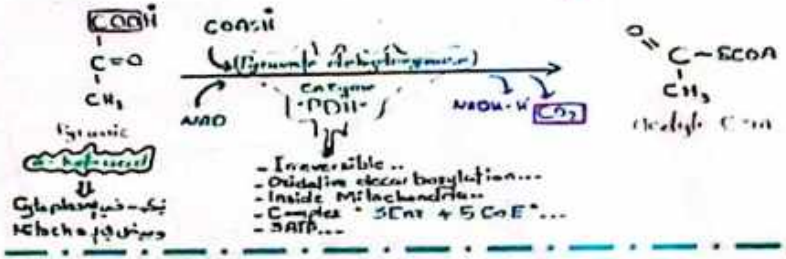

Malate Aspartate shuttle
AST
αKG
Glutamate
Malate dehydrogenase
NAD+
NADH+H+
Cytosol
Aspartic acid
Oxalo acetate
Malate
Inner Mit. Memb.
MD
Malate dehydrogenase
Mit. Matrix
3ATP
NADH+H+
NAD+


Glycerol Phosphate shuttle
Glycerol 3-P dehydrogenase
NADH+H+
NAD+
Dihydroxyacetone P
DHAP
Glycerol 3-P
outer
Inner
Glycerol 3-P dehydrogenase
FAD
2ATP
FADH2

Oxidative decarboxylation of pyruvate

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- **pyruvate dehydrogenase complex (PDC or PDH) is formed of:**
  - ✓ **3 enzymes** (pyruvic dehydrogenase, dihydrolipoic transacetylase & dihydrolipoic dehydrogenase).
  - ✓ **5 coenzymes** (Thiamine PP, lipoic acid, CoASH, FAD, NAD).



Regulation of Pyruvate dehydrogenase complex

1. Allosteric regulation

- **Stimulated** by Pyruvate, ADP & NAD.
- **Inhibited** by its products acetyl CoA and NADH.

2. Covalent modification

- **Active by Dephosphorylation by pyruvate dehydrogenase phosphatase. "Insulin favors active form".**
  - ✓ **PDC phosphatase** require **calcium** for full activity.
  - ✓ (In the heart, increased intra-mitochondrial Ca^{2+} during rapid contraction activates the phosphatase, thereby increasing the amount of active, non-phosphorylated PDC).

- **Inhibited by phosphorylation by pyruvate dehydrogenase Kinase. "Glucagon favors inactive form".**
  - ✓ **PDC kinase** is, itself, inhibited allosterically by **ADP and pyruvate.**

    Stimulators of PDH inhibit PDH kinase.
  - ✓ **PDC kinase** is, itself, activated allosterically by **acetyl CoA and NADH.**

    Inhibitors of PDH activate PDH kinase.

Abnormalities of the PDC

1. Congenital deficiency of the enzyme

- ✓ rare

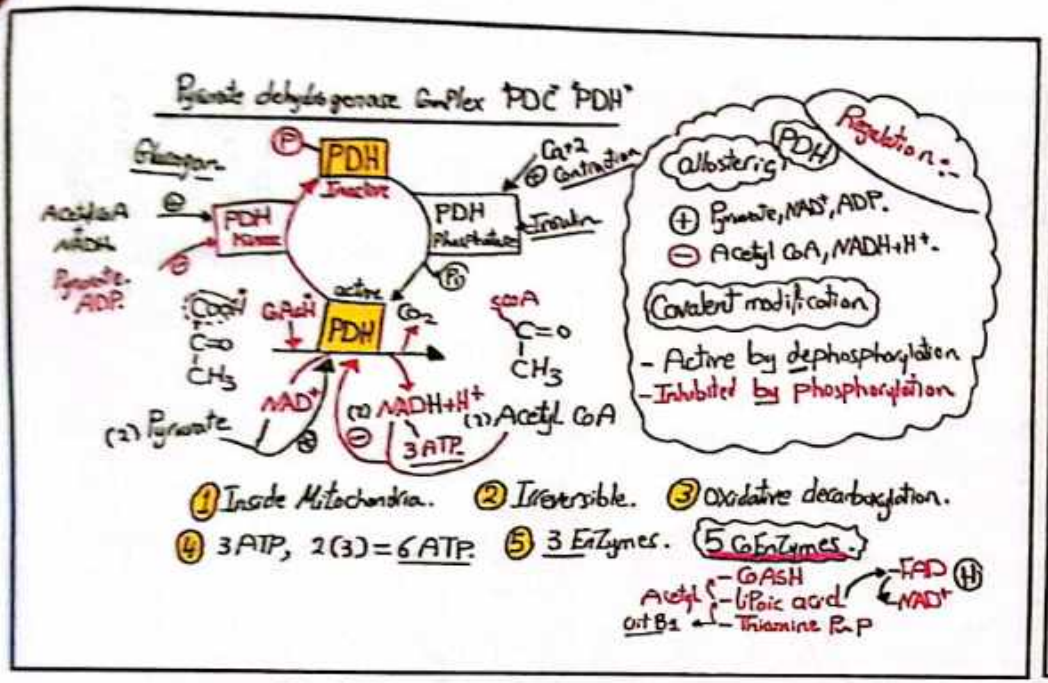
2. Dietary deficiency of thiamine pyrophosphate (TPP) in populations that use polished rice (deficiency of Vitamin B1)

- ✓ **Beriberi.**
- ✓ Brain is a prominent tissue where these metabolic defects manifest themselves leading to defective neural functions **(Wernicke-Korsakoff syndrome).**

3. Alcohol addiction is accompanied by nutritional deprivation of thiamin pyrophosphate.

4. Arsenite: **Inhibits** enzymes require **lipoic acid** as a cofactor as PDC and α- ketoglutarate dehydrogenase complex (a TCA cycle enzyme).

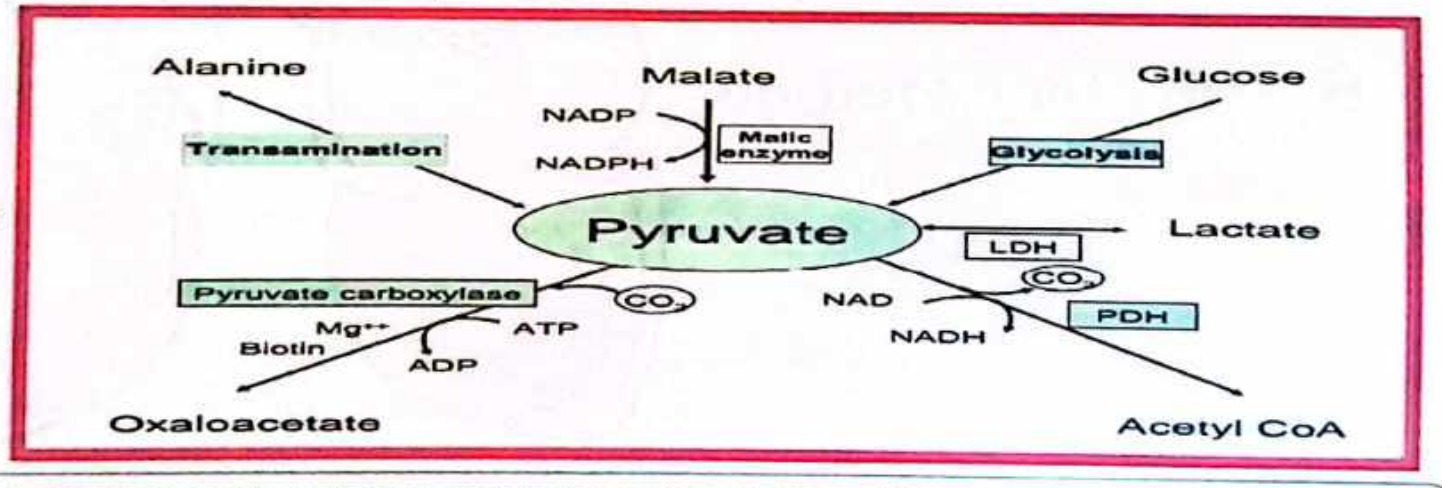
- ✓ **Pyruvate & lactate accumulate.**



Important Notes:

- **Phase 2 of cellular respiration:**
  - ✓ The energy derived from fuel oxidation is converted to the high-energy phosphate bonds or ATP by the process or **oxidative phosphorylation.**
- Some individuals with **alcoholism** develop **Wernicke-Korsakoff syndrome** because brain depends on glucose as a preferred fuel, so it is a prominent tissue where these metabolic defects manifest themselves.
- **Abnormalities** in PDC leads to **increase lactic acid** in blood and decrease pH, this serious condition called **lactic acidosis.**

Other Fates of Pyruvate:



AutoCross

Pyruvate carboxylase and PDC catalyze irreversible reactions. While LDH and transamination reactions are reversible.

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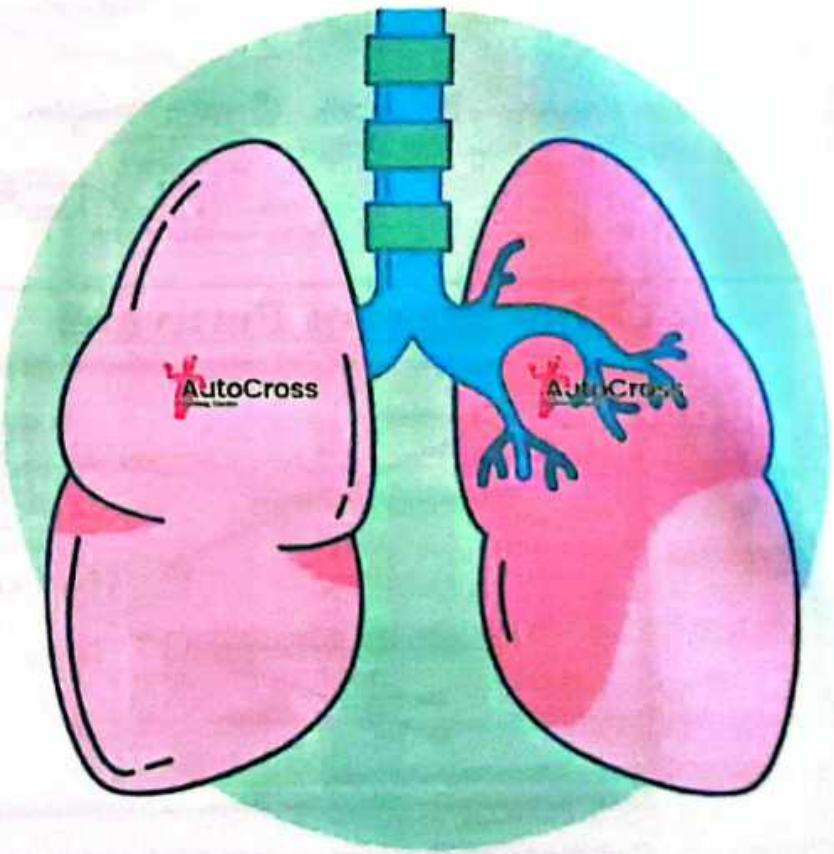
BIOCHEMISTRY ACADEMY

Chapter

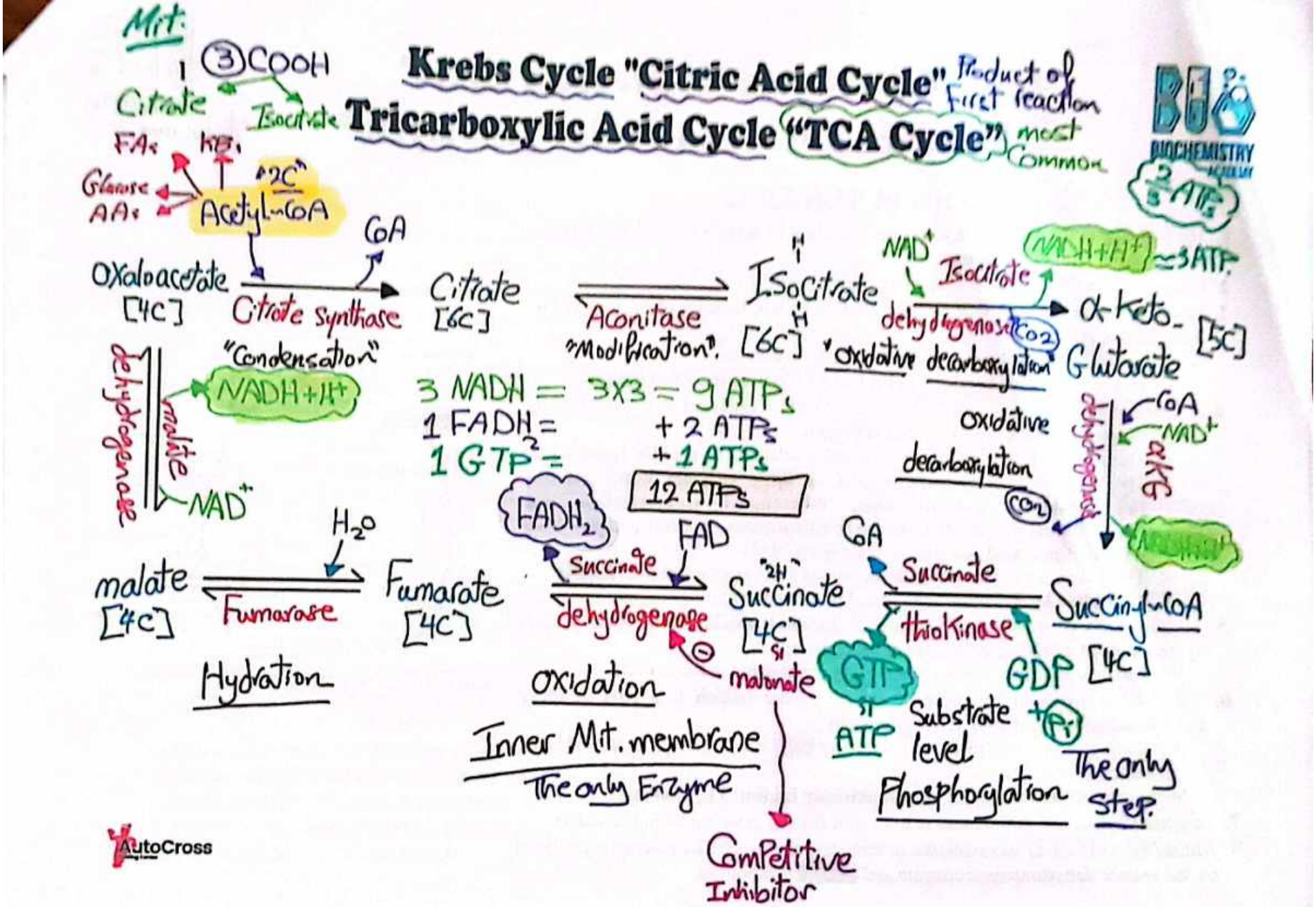
Krebs Cycle &

Lung Diseases

Respiratory Module



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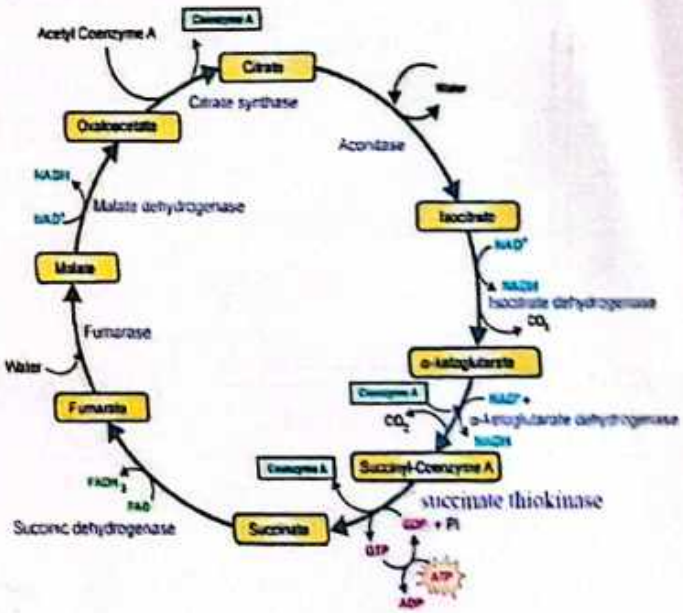

Mit.
Krebs Cycle "Citric Acid Cycle"
Tricarboxylic Acid Cycle "TCA Cycle"
Product of First reaction most Common
BIOCHEMISTRY
2.5 ATPs
1 ③COOH
Citrate
Isocitrate
FAs
KBs
Glucose
AAs
"2C"
Acetyl-CoA
CoA
Oxaloacetate [4C]
Citrate synthase
"Condensation"
Citrate [6C]
Aconitase
"Modification"
IsoCitrate [6C]
H
NAD+
Isocitrate dehydrogenase
(NADH+H+) ≈3ATP
CO2
"oxidative decarboxylation"
α-keto-Glutarate [5C]
dehydrogenase
malate
NAD+
NADH+H+
3 NADH = 3X3 = 9 ATPs
1 FADH2 = + 2 ATPs
1 GTP = + 1 ATPs
12 ATPs
oxidative decarboxylation
CO2
dehydrogenase
CoA
NAD+
αKG
NADH+H+
FADH2
FAD
H2O
malate [4C]
Fumarase
Fumarate [4C]
Succinate dehydrogenase
"2H"
Succinate [4C]
malonate
CoA
Succinate thiokinase
GTP
ATP
GDP
Pi
Succinyl-CoA [4C]
Substrate level Phosphorylation
The only step
Hydration
oxidation
Inner Mit. membrane
The only Enzyme
Competitive Inhibitor
AutoCross

Krebs Cycle "Citric Acid Cycle"
Tricarboxylic Acid Cycle "TCA Cycle"

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Reactions of TCA Cycle

1. The first step in the TCA cycle is the **condensation** of acetyl CoA and oxaloacetate leading to the formation of citrate.
2. Citrate molecule is **modified** to form isocitrate.
3. Isocitrate undergoes **oxidation decarboxylation** by isocitrate dehydrogenase and α-ketoglutarate is formed.
   - ✓ isocitrate dehydrogenase is a NAD-dependent enzyme.
   - ✓ This reaction leads to release of the first CO_2 and formation of **NADH**.
4. α -ketoglutarate undergoes **oxidative decarboxylation** catalyzed by α -ketoglutarate dehydrogenase complex. Succinyl CoA is formed.
   - ✓ This reaction is very similar to the reaction catalyzed by the PDC (pre-Krebs).
   - ✓ The enzyme is a complex formed of **three enzymes** bound together **(α-ketoglutarate dehydrogenase, dihyrolipoyl transacetylase and dihydrolipoyl dehydrogenase)** and **five coenzymes** (thiamine pyrophosphate = (TPP), lipoic acid , coenzyme A, FAD and NAD+).
   - ✓ This reaction leads to release of the second CO_2. and formation of **NADH**.
   - ✓ **"Inhibited by Arsenite** "inhibitor of **lipoic acid** requiring enzymes"
5. Succinyl CoA is converted to succinate by **succinate thiokinase** (or succinyl CoA synthetase that catalyzes the reverse of the reaction).
   - ✓ The only **substrate level phosphorylation** reaction in Krebs cycle.
6. Succinate is **oxidized** by succinate dehydrogenase **(which is a part of inner mitochondrial membrane)** to form Fumarate.
   - ✓ The is the only step in TCA cycle where **FAD** is the cofactor for the oxidation reaction.
   - ✓ Succinate dehydrogenase is **competitively Inhibited by Malonate.**
7. Fumarate by the action fumarase is **hydrated** leading to the formation of malate.
8. Malate is **oxidized** to oxaloacetate, to complete the cycle. The reaction is catalyzed by the malate dehydrogenase enzyme and **NADH** is produced.



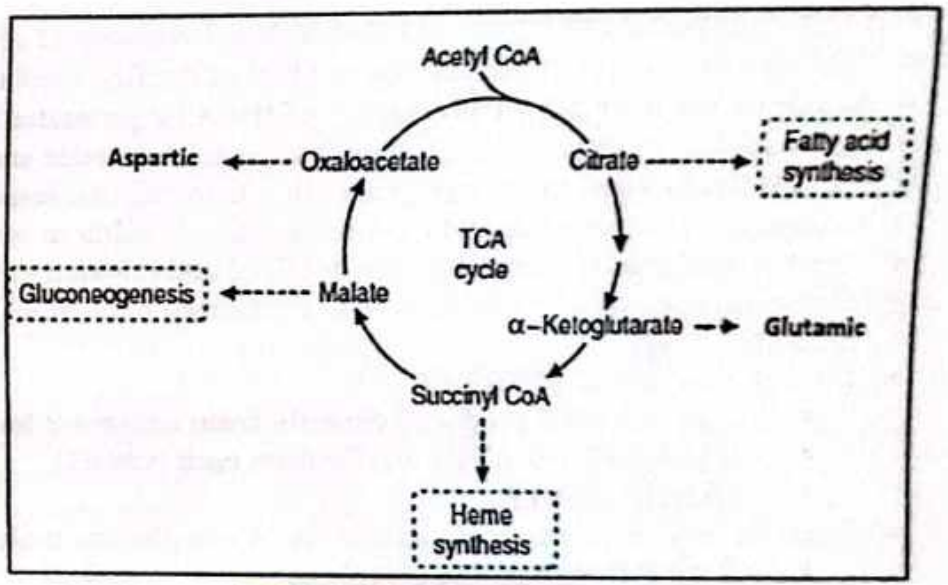
Krebs Cycle "Citric Acid Cycle"
Tricarboxylic Acid Cycle "TCA Cycle"

- ✓ **Tricarboxylic Acid Cycle (TCA-Cycle):** Involvement of tricarboxylates "Citrate, Isocitrate" **"The most common name".**
- ✓ **Citric acid cycle:** Citrate was the product of the first reaction of TCA.
- ✓ **Accounts for over 2/3 "Two thirds" of the ATP** generated from fuel oxidation. **"Occur exclusively in Mitochondria".**
- ✓ For oxidation of fatty acids, glucose, amino acids, acetate and ketone bodies "all generate acetyl CoA".
- ✓ **TCA is the central to energy generation** from cellular respiration.
- ✓ Complete oxidation of acetyl CoA inside mitochondria to produce **two CO_2, three $NADH+H^+$, one $FADH_2$ & one GTP.**
- ✓ The two carbons of acetyl CoA are oxidized and released in the form of **2 CO_2**
- ✓ Occur **only in aerobic conditions** to reoxidized back 3 $NADH+H^+$ & one $FADH_2$ into NAD & FAD by respiratory chain that need O_2.
- ✓ **Energy produced: (12 ATP)**
  - **1 GTP = 1 ATP** produced **directly from substrate level phosphorylation by Succinate Thiokinase.**
  - **3 $NADH+H^+$ = 9 ATP** (3 ATP from each NADH).
  - **1 $FADH_2$ = 2 ATP.**
- ✓ **Total energy from complete oxidation of one glucose** molecule under **aerobic conditions** will yield **a net of 36- 38 ATP**
  - **6- 8 ATP** from Aerobic glycolysis
  - **6 ATP** form Oxidative decarboxylation of pyruvate to acetyl CoA "PDH complex enzyme"
  - **(12 X 2) =24 ATP** Krebs' Cycle for 2 Acetyl CoA .
- ✓ **Oxidative decarboxylation reaction:**
  - Isocitrate dehydrogenase **"not complex".**
  - α ketoglutarate dehydrogenase (as PDH complex) need 3 Enzymes & 5 coenzymes.
- ✓ **Substrate level phosphorylation:**
  - Is the formation of a high-energy phosphate bond where none previously existed without the use of molecular O_2 or the ETC (in other words, it is NOT an oxidative phosphorylation). Occur only by **Succinate thiokinase in TCA Cycle.**
- ✓ **Four Members of the Vitamin B complex pay role in TCA Cycle "in the form of Coenzymes"**
  - **Niacin** in the form of NAD^+ , **Riboflavin** in the form of FAD, **Thiamin** (Vitamin B1) in the form of TPP, **Pantothenic acid** in the form of Coenzyme A.

Amphibolic aspects of TCA cycle

Definition:

- TCA can act as **catabolic or anabolic pathway.**



فكرني أقولك حاجتين مهمين جدا

1- Catabolic role:

✓ TCA is **the final common pathway** for oxidation of CHO , lipids & protein.

2- Anabolic role:

a) Synthesis of fatty acids:

✓ **After high CHO meal,** Acetyl CoA is converted to **Citrate**, efflux to cytoplasm & hydrolyzed to give acetyl CoA in the cytoplasm for synthesis of Fatty acids & Lipogenesis.

b) Non-essential amino acids:

✓ By transamination

- **Oxaloacetate ↔ aspartate.**
- **α-ketoglutaric ↔ glutamate.**

c) Synthesis of heme in liver and bone marrow:

- **Succinyl CoA** + glycine → δ amino levulinic acids.

d) Gluconeogenesis "glucose synthesis":

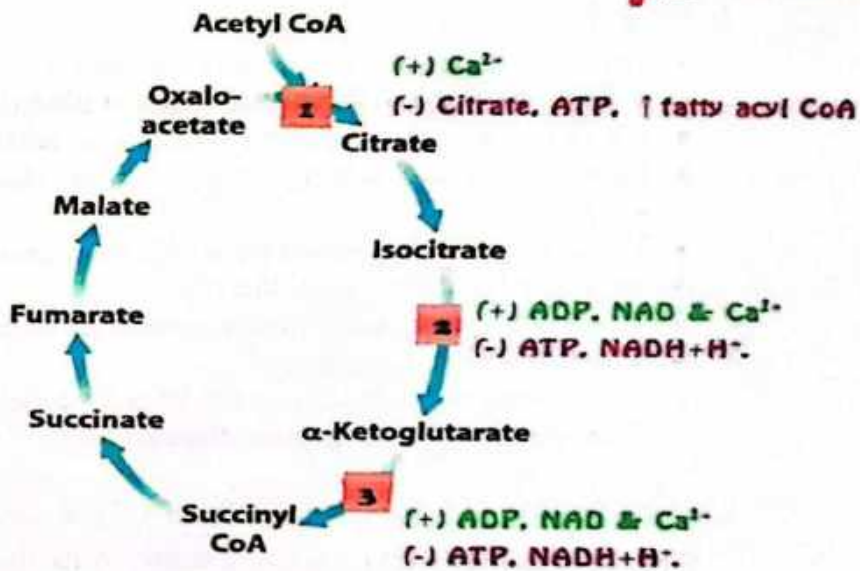
- **During fasting, Malate efflux** to cytoplasm to give glucose.

Regulation of Krebs' Cycle

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1-Allosteric regulation: "Fine"

1. Citrate synthase
2. Isocitrate dehydrogenase
3. α-ketoglutaric dehydrogenase".



2-Coarse control:

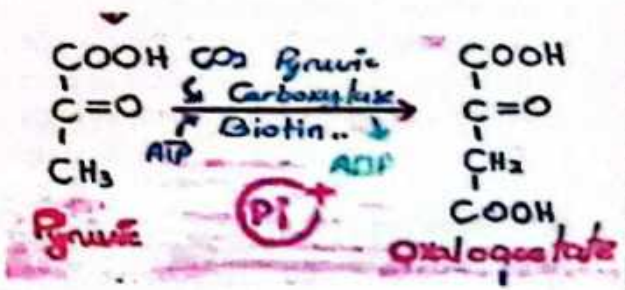
- The main function of the TCA cycle is energy production, which depends mainly on electron transport chain.
- Availability of oxygen and the concentration of ADP in the cell constitute the coarse control of the cycle. So the rate of utilization of ATP determines both the rate of respiration and the activity of the TCA cycle.

Anaplerotic Reactions:

Pathways or reactions that **replenish the intermediates** of the TCA cycle are referred to as anaplerotic reactions ("filling up").

Pyruvate carboxylase is a major anaplerotic enzyme:

- ✓ Found in liver, Brain, adipocytes and Fibroblasts
- ✓ Pyruvate carboxylase is **activated by acetyl CoA** (When acetyl CoA accumulates. it allosterically stimulates pyruvate carboxylase to ensure proper concentration of oxaloacetate for the oxidation of the excess acetyl CoA).
- ✓



N.B.: **Four members of the vitamins B complex** used as coenzymes in Krebs' cycle:

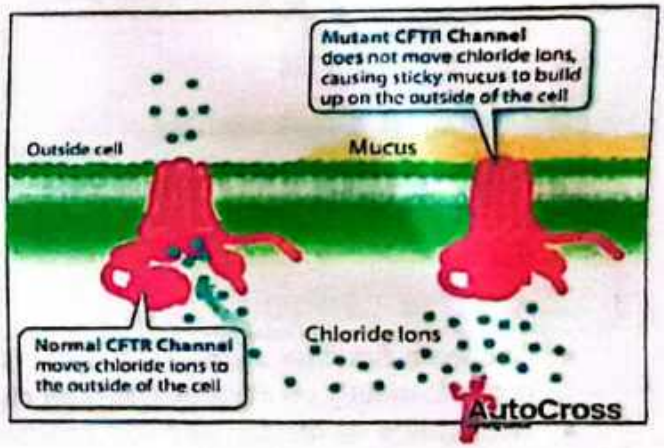
- **Vitamin B1 "Thiamine", Niacin, Riboflavin & Pantothenic acid.**

Molecular Basis of Lung Diseases

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Cystic Fibrosis (CF):

- Caused by mutations to the gene for the **CF transmembrane regulator (CFTR)** **"deletion of three nucleotides of phenylalanine"**.
- (CFTR) protein act as a **chloride channel** on epithelium of Pancreas, lungs.
- Defective CFTR, decreased secretion of chloride → **thick, viscous secretions.**
- **Thick mucus with obstruction of GIT & respiratory tract.**
- Increase chance of bacterial **infection.**
- **Pancreatic enzymes** are not able to reach the intestine. **"Digestive problem"**
- **Life threatening condition.**
  - ✓ Death occurs usually by the 2nd or 3rd decades.
- Most common Serious **genetic disease** in **UK.**



Emphysema α1 antitrypsin deficiency:

- α_1-antitrypsin (α_1,-AT) is secreted mainly by the liver, monocyte & alveolar macrophage.
- It **inhibits a number of proteolytic enzymes (elastase)** that is **secreted by the neutrophils** and degrades elastin of alveolar walls if the lungs.
- The **α1 antitrypsin deficiency** is a genetic defect **alters the configuration of the alpha1-antitrypsin molecule and prevents its release from hepatocytes.**
- As a result, serum levels of alpha1-antitrypsin are decreased, leading to low alveolar concentrations, where the alpha1-antitrypsin molecule normally would serve as protection against proteases such neutrophils elastase. The excess elastase in alveoli **destroys alveolar walls and causes emphysema (no elastic recoil).**
- The accumulation of excess mutated alpha1-antitrypsin in hepatocytes can also lead to destruction of these cells and **liver disease.**

N.B.: Smoking adds to the severity of the disease. Smoking causes **oxidation and subsequent inactivation of methionine in α1-antitrypsin (Methionine 358** in a1- antitrypsin is required for the binding of a1- antitrypsin to its target elastase). So, **α1- antitrypsin can't bind to elastase.** This leads to lung **emphysema.**